

# Project Introduction

Arjun

## Project Introduction

The **Document Signature App** is a secure, full-stack web application that enables users to **upload documents, place digital signatures, share signing links, and generate legally traceable signed PDFs** — similar to platforms like **DocuSign** and **Adobe Sign**.

The app eliminates the need for physical paperwork by allowing documents to be signed electronically with full **audit trails, signer identity verification, and document integrity**. It is designed with real-world enterprise workflows in mind, including authentication, document ownership, status tracking, and signature history.

This project demonstrates how modern SaaS products handle **file security, digital trust, and collaborative workflows** at scale.

## Aim of the Project

## Aim of the Project

The primary aim of this project is to:

- Build a **production-ready digital signature system**
- Understand **end-to-end document lifecycle management**
- Learn how **secure SaaS platforms** are architected
- Simulate real **enterprise-grade backend and frontend workflows**
- Gain hands-on experience with **PDF processing, audit logging, and token-based access**

This project is not just a CRUD app — it models **real business logic** used in legal, HR, finance, and enterprise software.

## Core Problems This App Solves

## Core Problems This App Solves

- Manual document signing is **slow and error-prone**
- Physical paperwork is **hard to track and store**
- Emailing PDFs back and forth has **no auditability**
- No visibility into **who signed, when, and from where**
- Risk of document tampering after signatures

This app addresses all of these with **secure uploads, signature coordinates, immutable signed PDFs, and audit trails.**



# Real-World Use Cases



# Real-World Use Cases

## 1 Business Contracts & Agreements

- Vendors sign contracts remotely
- Status tracked as *Pending / Signed / Rejected*
- Signed documents stored securely

## 2 HR & Onboarding

- Offer letters
- NDA signing
- Policy acknowledgements

## 3 Freelancers & Agencies

- Client agreements
- Proposal approvals
- Payment authorization documents

## 4 Legal & Compliance Teams

- Legal documents with traceability
- Audit trails for disputes
- IP-based signer tracking

## 5 Education & Institutions

- Consent forms



- Admission documents
- Certificates and approvals

## Key Features That Mimic Industry Standards



## Key Features That Mimic Industry Standards

- JWT-based authentication
- Secure PDF upload & access control
- Drag-and-drop signature placement
- Server-side PDF modification (PDF-Lib)
- Tokenized public signature links
- Audit logs with timestamps & IP
- Status lifecycle: Pending → Signed → Rejected
- Deployment-ready architecture

These are the **exact patterns used in enterprise SaaS products**.



## Industry Value of This Project



## Industry Value of This Project



## Why Companies Care About This Project

This project showcases **skills companies actively hire for**:



### SaaS Architecture

- Multi-user authentication
- Role-based document ownership
- Tokenized external access



### Security & Compliance Thinking

- JWT authentication
- Audit trail implementation
- Protected document access
- Immutable signed outputs



### Backend Engineering Depth

- File handling at scale
- PDF manipulation
- Middleware logging
- Status workflows



### Frontend Product Thinking

- Drag-and-drop UX
- PDF rendering

- Responsive dashboards
- Status-based UI filtering

### **Real Business Logic (Not Tutorials)**

- Document lifecycle management
  - Signature coordinates mapping
  - Final PDF generation
  - Email/share workflows
-

# What This Project Signals on Your Resume

## What This Project Signals on Your Resume

- ✓ “This developer can build **enterprise-grade SaaS features**”
- ✓ “Understands **security, audit, and compliance basics**”
- ✓ “Can work with **files, PDFs, and real data flows**”
- ✓ “Not just a frontend/backend dev — but a **product-aware engineer**”

This project sits in the same category as:

- CRM systems
  - HR platforms
  - LegalTech software
  - FinTech document systems
- 

## Why This Is a Strong Portfolio Project

DocuSign-style apps prove you can handle real-world complexity beyond CRUD apps

Compared to todo apps or basic dashboards, this project demonstrates:

- **Stateful workflows**
- **Security-first design**
- **Cross-user interactions**
- **Document integrity guarantees**

It's the kind of project that **stands out immediately** in interviews.





## 2-Week Build Plan



## 2-Week Build Plan



### Week 1: Core Features, Backend & Frontend Setup

#### Day 1: Project Setup & Repo Initialization

- Create GitHub repo with MERN folder structure
- Initialize React app with Tailwind CSS
- Setup Node.js + Express + MongoDB (Mongoose)
- Install necessary libraries (Multer, PDF-Lib, bcrypt, JWT)  
**Skills:** Project scaffolding, Tailwind setup, server setup

#### Day 2: Auth System (JWT)

- User model: name, email, password
- Routes: /register, /login
- JWT auth + bcrypt password hashing
- Auth middleware for protected routes  
**Skills:** JWT, secure user auth, API routes

#### Day 3: File Upload API

- Setup Multer for PDF uploads
- Store file path & metadata in MongoDB
- Route: /api/docs/upload  
**Skills:** File handling, middleware, API design

#### Day 4: View & List Documents

- API to fetch user's uploaded files
- Display files in a dashboard
- Add preview functionality using `react-pdf`  
**Skills:** RESTful APIs, PDF rendering in React

#### Day 5: Signature Schema & Logic

- Signature Model: field, coordinates, signer, status
- Route to save signature positions (x, y)
- Display signature placeholder on PDF  
**Skills:** Schema relations, position-based rendering

#### Day 6: PDF Editor Integration

- Add drag-and-drop signature field (HTML overlay)
- Save coordinates relative to page
- Route: POST /api/signatures  
**Skills:** Coordinate systems, frontend UX logic

#### Day 7: Buffer / Testing

- Debug UI and backend integration
  - Write Postman tests for API
-

## ✓ Week 2: Signature Rendering, Sharing & Polish

### Day 8: Generate Final Signed PDF

- Use **PDF-Lib** to embed signature text/image
- Export signed version to disk/cloud  
**Skills:** PDF processing, server-side document generation

### Day 9: Email + Public Signature Links

- Generate tokenized URL for external signature
- Email link to signer (use nodemailer or mock)  
**Skills:** Token auth, email logic

### Day 10: Audit Trail

- Log who signed, when, and IP (use middleware)
- Route: GET /api/audit/:fileId  
**Skills:** Middleware logging, basic audit management

### Day 11: Signature Status Updates

- Status: Pending, Signed, Rejected
- Allow signer to accept/reject with reason  
**Skills:** Status flows, conditional logic

### Day 12: Dashboard UI Polish

- Filter by signature status
- Responsive layout with Tailwind  
**Skills:** Tailwind UI polish, component reuse

### **Day 13: Deployment**

- Backend: Render / Railway
- Frontend: Vercel / Netlify
- MongoDB Atlas for DB

### **Day 14: Final Testing + Demo**

- Prepare a GitHub README
- Record a 2-min demo walkthrough

# Web Tech Stack for Document Signature App

# **Web Tech Stack for Document Signature App**

## **Recommended Tech Stack (Modern & Industry-Ready)**

### **Frontend**

- **React (Vite or Next.js)**
- **TypeScript** (highly recommended)
- **Tailwind CSS**
- **react-pdf**
- **dnd-kit** (drag & drop)
- **Axios**
- **React Hook Form**
- **Zod** (validation)

## Backend (API Layer)

- **Node.js + Express**

## Auth

- **JWT (access + refresh tokens)**
- **Protected routes**

## Storage

- **Local storage (dev)**
- **Supabase Storage / AWS S3 (prod)**

## Deployment

- **Frontend:** Vercel / Netlify
- **Backend:** Render / Railway
- **DB:** Supabase / MongoDB Atlas



# API Endpoints & Mock Data

## API Endpoints & Mock Data

### Auth APIs

POST /api/auth/register

POST /api/auth/login

### Document APIs

POST /api/docs/upload // Upload PDF

GET /api/docs/ // List user PDFs

GET /api/docs/:id // View specific doc

### Signature APIs

POST /api/signatures // Save signature position

GET /api/signatures/:id // Get signatures for document

POST /api/signatures/finalize // Embed signature into PDF

### Audit APIs

GET /api/audit/:docId

### Mock Data: **User**

```
{
  "name": "Alice",
  "email": "alice@example.com",
  "password": "hashedpassword"
}
```

### Mock Data: **Signature**

```
{
  "documentId": "abc123",
  "userId": "xyz456",
  "x": 120,
  "y": 330,
  "page": 1,
  "status": "signed"
}
```



## Skills You Will Learn by Stage



## Skills You Will Learn by Stage

| Week | Skill          | Description                             |
|------|----------------|-----------------------------------------|
| 1    | Auth & API     | JWT, bcrypt, REST APIs                  |
| 1    | File Upload    | Multer, server-side storage             |
| 1-2  | Frontend UX    | Drag-and-drop UI, react-pdf             |
| 2    | PDF Generation | <code>pdf-lib</code> for embedding data |
| 2    | Security       | Tokenized links, protected routes       |
| 2    | Audit Logging  | Middleware, action tracking             |
| 2    | Deployment     | Hosting full-stack app                  |

# Resources

## Learning Resources

1. PDF-Lib JS Docs: <https://pdf-lib.js.org/docs>
  2. MERN Auth (JWT): <https://www.youtube.com/watch?v=2jqok-Wgell> (PedroTech)
  3. Multer File Upload: <https://www.youtube.com/watch?v=9Qzmri1WaaE>
  4. PDF Signature with Node:  Signing a PDF with Javascript - Going though the code
  5. React PDF Viewer:  How to View PDFs in React JS with React PDF Viewer
  6. React Drag and Drop Signatures:  
 React Drag And Drop (dnd-kit) | Beginners Tutorial
- 

## Web Tech Stack Resources (Step-by-Step)

---

### 1. React (Core)

 YouTube

**React Full Course (Beginner → Advanced)**

 <https://www.youtube.com/watch?v=bMknfKXIFA8>

 **Covers:**

- Hooks
- State
- Components
- API calls

---

 Docs

- React Official Docs  
👉 <https://react.dev/>

---

## ⚡ 2. Vite / Next.js Setup

 YouTube

### Vite + React Setup

👉 [https://www.youtube.com/watch?v=GgMbL0z\\_h8A](https://www.youtube.com/watch?v=GgMbL0z_h8A)

### Next.js App Router (Optional)

👉 <https://www.youtube.com/watch?v=wm5gMKuwSYk>

---

## 🎨 3. Tailwind CSS (UI Polish)

 YouTube

### Tailwind CSS Crash Course

👉 <https://www.youtube.com/watch?v=dFgzHOX84xQ>

✅ Perfect for dashboards

---

## Docs

- **Tailwind Docs**

👉 <https://tailwindcss.com/docs>

---

## 4. PDF Rendering (CRITICAL)

### YouTube

**View PDF in React using react-pdf**

👉 <https://www.youtube.com/watch?v=Uu4W8r9sKxQ>

---

## Docs

- **react-pdf Docs**

👉 <https://github.com/wojtekmaj/react-pdf>

---

## 5. Drag-and-Drop Signatures (CORE FEATURE)

### YouTube

**Drag & Drop using dnd-kit**

👉 <https://www.youtube.com/watch?v=FZt2N6R0YkY>

---

## Docs



- dnd-kit Docs

👉 <https://docs.dndkit.com/>

---

## 6. API Integration (Axios + JWT)

 YouTube

### Axios Crash Course

👉 <https://www.youtube.com/watch?v=6LyagkoRWYA>

### JWT Auth in React

👉 <https://www.youtube.com/watch?v=nDGA3km5He4>

---

 Docs

- Axios Docs

👉 <https://axios-http.com/>

---

## 7. Auth & Protected Routes

 YouTube

### Protected Routes in React

👉 <https://www.youtube.com/watch?v=2k8NleFjG7I>

---

## 8. Forms & Validation

 YouTube

## React Hook Form + Zod

👉 <https://www.youtube.com/watch?v=Z1Nw5FfKqC8>

---

 Docs

- React Hook Form  
👉 <https://react-hook-form.com/>
  - Zod  
👉 <https://zod.dev/>
- 

## 9. Coordinate System (IMPORTANT)

Concept to Learn

- Convert absolute → relative coordinates
- Save (x %, y %)
- Re-render correctly on different screens

 Video

## Mouse Position & Bounding Box in React

👉 <https://www.youtube.com/watch?v=qd5y8T7Zx9U>

---

## 10. Deployment

 YouTube

## Deploy React App on Vercel

👉 <https://www.youtube.com/watch?v=2h6n5Dgk0oI>

---

### 🏆 Recommended Learning Order

| St<br>ep | Learn |
|----------|-------|
|----------|-------|

|   |                 |
|---|-----------------|
| 1 | React<br>basics |
|---|-----------------|

|   |          |
|---|----------|
| 2 | Tailwind |
|---|----------|

|   |           |
|---|-----------|
| 3 | react-pdf |
|---|-----------|

|   |                |
|---|----------------|
| 4 | Drag &<br>Drop |
|---|----------------|

|   |                |
|---|----------------|
| 5 | Axios +<br>JWT |
|---|----------------|

|   |                     |
|---|---------------------|
| 6 | Protected<br>routes |
|---|---------------------|

**7    Coordinat  
     e logic**

**8    UI polish**

**9    Deploym  
     ent**

---

 **Why This Web Stack Is Perfect**

**This tech stack matches real SaaS dashboards used in industry**

**You'll be learning:**

- **Component-driven UI**
- **Drag-and-drop UX**
- **Secure frontend auth**
- **PDF rendering workflows**
- **Responsive enterprise UI**

